



Socioeconomic and climatic impacts on long-term electricity demand: A high-resolution approach through machine learning

Jin Huang^a, Gregorio Iglesias^{a,b},*

^a MaREI, Environmental Research Institute & School of Engineering, College Road, Cork, P43 C573, Ireland

^b School of Engineering, Computing and Mathematics, Drake Circus, Plymouth, PL4 8AA, UK

ARTICLE INFO

Keywords:

Socioeconomic impact
Climate change scenarios
Long-term electricity demand prediction
Machine learning modelling
High-resolution electricity demand modelling

ABSTRACT

Reliable long-term electricity demand prediction is essential for strategic energy planning, particularly as nations transition to renewable energy systems. This study investigates the socioeconomic and climatic impacts on Ireland's long-term electricity demand using a high-resolution machine learning modelling approach. An artificial neural network (ANN) model is presented to forecast hourly electricity demand up to 2060 under various demographic, economic, and climatic scenarios. Applying real-world and publicly accessible datasets, the research examines the causal factors influencing variations in historical electricity demand across multiple temporal scales. The ANN model, optimized through advanced hyperparameter tuning, incorporates key drivers of electricity consumption, including population growth, GDP, temperature fluctuations, and behavioural patterns. Results reveal a persistent annual increase in electricity demand, driven primarily by demographic trends and economic growth, while different climate scenarios illustrate the impact of warming and extreme cold temperatures on demand profiles. The proposed AI-based approach offers researchers, energy planners, and policymakers a simple and robust tool for modelling high-resolution energy systems and supporting the alignment of renewable energy targets with future consumption needs.

1. Introduction

Electricity is the backbone of modern society, driving economic growth, technological progress, and improvements in quality of life. At the same time, social and economic development significantly influences electricity consumption patterns. As a critical resource, its uninterrupted availability is essential to meet the needs of households, industries, and public infrastructure. Therefore, accurate prediction of electricity demand is pivotal in ensuring energy system stability, guiding energy planning and resource allocation [1–4]. In addition, many countries are adopting ambitious decarbonization targets, transitioning from fossil fuels to renewable energy [5–8]. In this context, understanding the socioeconomic and climatic impacts on future demand is key to aligning renewable energy development with carbon reduction goals, while maintaining energy security [9–12].

Electricity demand forecasting is typically categorized based on time horizons into short-term, medium-term, and long-term forecasts. Each type serves distinct purposes, with different timeframes catering to specific operational and planning needs of utility companies. Short-term forecasts [e.g. 3,4,9,11,13–15], which span hours to a few days, are essential for real-time grid operation and balancing supply with fluctuating demand. Medium-term forecasts [e.g. 15–19], covering

weeks to months, aid in resource scheduling and maintenance planning. Long-term forecasts [e.g. 10,15,20–22], spanning years to decades, are crucial for strategic energy planning, infrastructure investment, and policy formulation. This study focuses on long-term electricity demand forecasting, aimed at supporting energy system planning and renewable energy strategies under a variety of socioeconomic and climate change scenarios.

However, electricity demand forecasting is a complex task [4,11,13,15,21–23]. Demand is shaped by a variety of factors, including population variation, economic activity, technological advancements, weather conditions, etc., each of which introduces different levels of uncertainty and variability. Moreover, consumption patterns exhibit considerable randomness, complicating efforts to model and predict demand accurately.

To this end, researchers have developed a variety of models to forecast electricity demand across different regions and countries. These models can be broadly classified into statistical-based approaches and machine learning techniques [1,11,17]. Statistical models, such as linear regression, auto-regressive integrated moving average, etc., are traditional and interpretable but often struggle to capture the non-linear and dynamic relationships that characterize electricity demand.

* Corresponding author at: MaREI, Environmental Research Institute & School of Engineering, College Road, Cork, P43 C573, Ireland.
E-mail address: gregorio.iglesias@ucc.ie (G. Iglesias).

By contrast, machine learning methods, including artificial neural networks (ANNs), support vector regression, grey systems, fuzzy logic, etc., have shown greater accuracy by effectively modelling these complex patterns [1,3,11,24]. As a result, machine learning approaches, particularly ANNs, have gained prominence in electricity demand forecasting research and will be employed in this study to forecast future electricity demand.

Long-term electricity demand forecasting plays a pivotal role in enabling policymakers to make informed planning decisions that align with the sustainable development goals of their countries. Accurate forecasts are essential for determining the size, location, and types of future power plants, as well as planning the routes and capacities of power transmission and distribution networks [2]. Previous studies [e.g. 25–28] have optimized the configuration of hybrid renewable energy systems – typically comprising wind turbines, solar photovoltaic panels, battery/hydrogen storage – based on historical electricity demand. While these configurations have the potential to achieve 100% clean energy production shortly, their long-term viability remains uncertain as social and economic conditions evolve. In other words, it is unclear whether such configurations will continue to support full decarbonization or remain the optimal choice under changing circumstances.

In the authors' previous study [12], the long-term electricity demand in Ireland was estimated to explore pathways to achieving Net Zero by 2050, based on daily energy indicators provided by the Copernicus Climate Change Service (CCCS) [29]. However, this database focused solely on climate factors and neglected other influential variables, such as socioeconomic factors. Consequently, the annual electricity demand, calculated by aggregating daily data, exhibited a nearly constant year-to-year pattern, which conflicted with the upward trend observed in Ireland's electricity consumption over the past decade [30]. This study, therefore, aims to overcome these limitations by developing a comprehensive, high-resolution artificial intelligence model that integrates both socioeconomic and climatic factors. By analysing historical electricity consumption trends across multiple temporal scales and identifying the primary drivers of demand, this research aims to deliver more reliable long-term forecasts.

Previous studies have explored medium- and long-term electricity demand forecasting at various temporal resolutions, achieving varying levels of accuracy. For yearly resolution, [20] applied an ANN model to historical electricity demand data in Thailand from 1986 to 2010, achieving a Mean Absolute Percentage Error (MAPE) as low as 0.996%. [10] investigated long-term electricity demand forecasting under different scenarios with varying shares of solar and wind energy in the energy mix. Their method, based on 30 years of load data, achieved MAPEs ranging from 1.01% to 1.86% across different input data types and economies. For monthly resolution, [16] employed a multiple linear regression model to analyse the effect of meteorological variability on Italy's monthly electricity demand. Their forecasts, which predicted monthly aggregated demand up to one month ahead, yielded MAPEs of approximately 1.30%. Additionally, [18] developed a hybrid approach that combined neural networks for trend prediction with Fourier series to model the periodic behaviour of Spanish electricity demand. Their method achieved a MAPE of 1.74% for the years 1998 to 2002. Likewise, [17] decomposed Thailand's electricity demand into trend and cyclical components, which were separately modelled using two neural networks. Their approach achieved a MAPE of 1.42% for a one-year forecast.

While previous studies have focused on medium- or long-term forecasts at daily [e.g. 15,19], monthly [e.g. 16–18,22] or even yearly resolution [e.g. 10,20,21], high-resolution energy system simulations [e.g. 31–35], often performed with software such as EnergyPLAN [36], PLEXOS [37], HOMER [38], H2RES [39], etc., require hourly or even finer temporal resolutions. These high-resolution forecasts enable detailed load curves, better simulating load fluctuations, peak electricity demand, unit commitment, energy storage needs, and grid dispatch at

different times [12]. However, existing energy planning studies [40–42] often relied on static or retrospective electricity demand estimates, with limited consideration for socioeconomic and climate variability. Mispredicting these trajectories can lead to increased investment and operational costs [1,2,10,11,22,23] and reduced system reliability. Therefore, reliable long-term forecasts are essential, yet studies at such high resolution, particularly those with hourly data, are still relatively rare in the literature.

To address this gap, this study presents high-resolution long-term electricity demand forecasting for Ireland up to 2060, integrating socioeconomic and climate change scenarios. Notably, [43] developed a Python package to generate 21st-century hourly electricity load profiles across the conterminous United States, capturing the impacts of climate and socioeconomic changes at both spatial and temporal scales. In the present study, a deep ANN is employed, trained on real-world datasets that encompass Ireland's demographic trends, economic indicators, temperature variations, and behavioural factors. The novelty of this work lies in its use of comprehensive scenario analysis with advanced machine learning techniques to deliver hourly forecasts of national electricity demand under varying population, GDP, and climate change trajectories. The findings aim to provide policymakers, energy planners, and researchers with an advanced tool for strategic energy system planning, supporting the alignment of renewable energy development with future demand needs.

The paper is organized as follows: Section 2 analyses historical electricity consumption variations across different time scales. Section 3 presents the machine learning methodology, including data processing, model training, validation, and testing. Section 4 describes the socioeconomic and climate change scenarios considered in this study. Section 5 presents the predicted results and their implications. Section 6 concludes and discusses the study. Ultimately, this research could provide a robust foundation for long-term renewable energy planning [12], particularly for countries that rely heavily on both renewable and conventional energy sources, such as Ireland [44].

2. Historical electricity consumption

Fig. 1 illustrates the hourly electricity consumption in Ireland for the entire year of 2023 (comprising 8760 h), with a subplot displaying the consumption pattern over a representative month. The figure shows significant fluctuations at monthly, weekly, and daily intervals. In this study, ANNs will be used to forecast future electricity demand. Therefore, the careful selection of explanatory input features is critical for training the neural networks and ensuring high-performance electricity demand forecasting [4,17,24]. In the subsequent subsections, we will analyse Ireland's historical hourly electricity consumption from 2015 to 2023 across different temporal scales and introduce the relevant influencing factors. This analysis serves two purposes: firstly, to account for the long-term development trends essential for accurate long-term forecasting; and secondly, to capture recurrent temporal patterns – such as daily, weekly, and seasonal cycles – that influence peak demand magnitudes and are critical for high-resolution predictions. Note that the data used in this study, along with detailed descriptions of the sources, are provided in [Appendix](#).

2.1. Annual variations

Fig. 2 presents the yearly statistics of electricity consumption in Ireland from 2015 to 2023, including average, maximum, and minimum values for each year. It should be noted that in the following the box plots are not intended to represent random variation, but rather to summarize distribution patterns in electricity demand (or temperature) in considered time intervals. Historical data from Ireland's electricity grid operator present a steady increase in electricity demand, with annual mean values rising from 3031 MW in 2015 to 3706 MW in 2023. These annual variations can primarily be attributed to social and

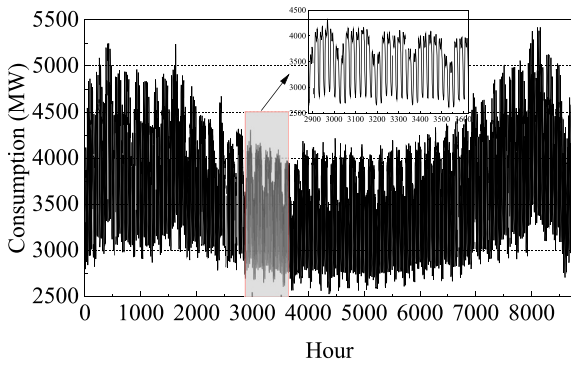


Fig. 1. Hourly electricity consumption in Ireland for 2023.

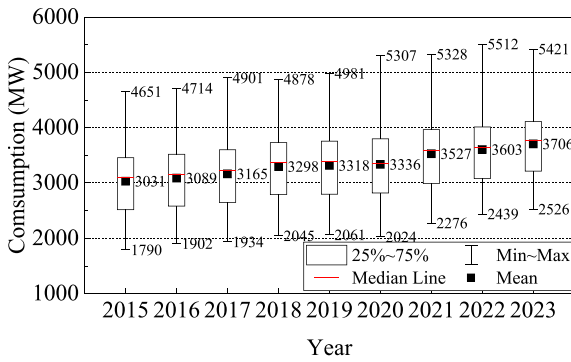


Fig. 2. Yearly statistics of electricity consumption in Ireland from 2015 to 2023.

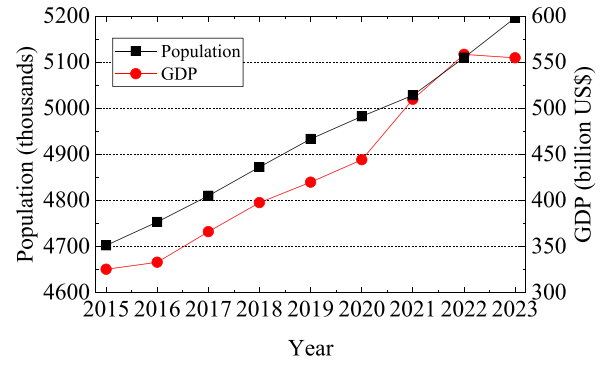


Fig. 3. Population and GDP in Ireland from 2015 to 2023.

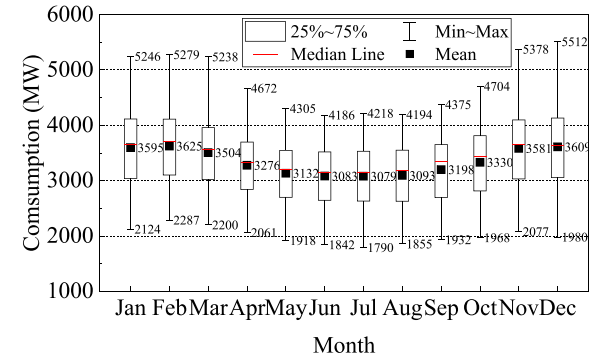


Fig. 4. Monthly statistics of electricity consumption in Ireland from 2015 to 2023.

economic developments [2,9,21–23], which influence both residential, commercial and industrial electricity consumption.

To quantify the impact of these factors, this study uses population and gross domestic product (GDP) as representative parameters for social and economic changes. Population directly affects electricity demand, as larger populations typically increase residential and commercial energy usage, particularly in densely populated urban areas. GDP, on the other hand, reflects industrial and commercial energy needs, with higher GDP levels indicating a more prosperous economy, often accompanied by greater industrial activity and, consequently, higher electricity consumption. As noted by [18], increasing wealth and technological advancements typically lead to rising electricity demand, although the growth rate may vary in response to economic fluctuations.

Fig. 3 illustrates the trends in population and GDP in Ireland from 2015 to 2023. During this period, Ireland's population grew from approximately 4.7 million to 5.2 million, while GDP expanded at an average annual rate of 6.9%. The Pearson correlation coefficients between electricity demand and population is ~ 0.978 , and a similarly high value is observed with GDP, indicating a strong positive relationship in both cases.

These findings align with previous studies. For instance, [45] reported a correlation coefficient of 0.938 between electricity use per capita and GDP per capita in Ireland from 1960 to 1995. Excluding the effect of population, the Pearson correlation coefficient between total electricity use and GDP in the same period was 0.990. These consistent results illustrate the strong linkage between electricity demand and socio-economic factors.

2.2. Monthly variations

Fig. 4 provides the monthly statistics of electricity consumption in Ireland from 2015 to 2023, including average, maximum, and minimum values in each month. The results indicate a distinct monthly or

seasonal pattern, with higher electricity demand in winter and lower demand in summer. This pattern is primarily influenced by Ireland's high-latitude location, where cold winters increase the need for heating in residential and commercial buildings. Many households and businesses rely on electric heating systems, such as electric heaters and heat pumps, to stay warm, thereby driving up electricity consumption during winter months. Conversely, Ireland's mild summer temperatures result in minimal demand for air conditioning, which is typically limited to certain public spaces. Consequently, climate plays a significant role in shaping the seasonal patterns of electricity consumption in Ireland.

Previous studies [e.g. 4,16,46] have shown the strong linkage between electricity consumption and meteorological factors such as temperature, humidity, wind speed, cloud cover, etc. These studies consistently demonstrate temperature as the dominant factor affecting electricity demand, though additional variables, such as relative humidity and cloud cover, can marginally improve prediction accuracy. Additionally, [16], which analysed electricity demand in Italy, indicated that the relationship between temperature and electricity demand is often non-linear, as both increases and decreases in temperature beyond certain thresholds can result in higher electricity usage, and for colder regions, temperature reductions during winter have a more pronounced impact on electricity consumption than temperature increases during summer. Accordingly, this study employs ANNs, which are well-suited to capturing non-linear relationships between variables, and uses temperature as the primary representative variable to assess the influence of the climate on Ireland's electricity demand.

The monthly variation in temperature from 2015 to 2023 in Ireland is shown in Fig. 5. A clear inverse trend is observed between electricity consumption and temperature, with a Pearson correlation coefficient of -0.935 , indicating a strong negative relationship. Furthermore, longer daylight hours in summer reduce the need for artificial lighting in the evenings and early mornings, contributing to lower electricity

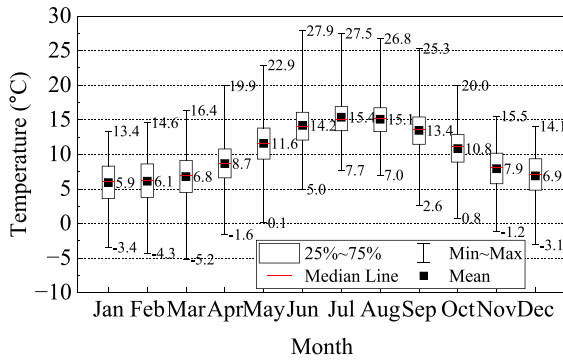


Fig. 5. Monthly statistics of temperature in Ireland from 2015 to 2023.

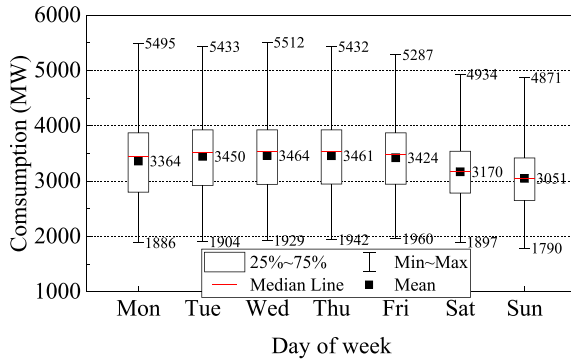


Fig. 6. Weekly statistics of electricity consumption in Ireland from 2015 to 2023.

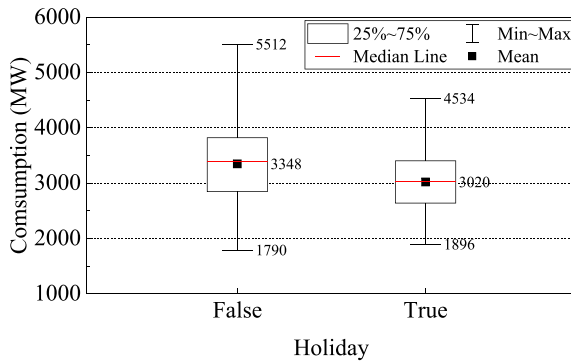


Fig. 7. Electricity consumption on holidays versus non-holidays in Ireland from 2015 to 2023.

consumption. A more detailed discussion of electricity consumption variations at different times of the day will be presented in Section 2.3.

2.3. Weekly and hourly variations

Fig. 6 shows the weekly statistics of electricity consumption in Ireland from 2015 to 2023. The data show that electricity demand is consistently higher from Monday to Friday compared with Saturday and Sunday. This trend reflects the increased energy use during weekdays when businesses, schools, and industries operate at full capacity. In contrast, weekends typically experience reduced commercial and industrial activity, resulting in lower electricity demand. Similarly, Fig. 7 illustrates the difference in electricity consumption on holidays versus non-holidays. Non-holidays generally see higher electricity demand due to the operation of workplaces and public services, whereas holidays are associated with a significant decline in energy use.

These findings align with prior studies [e.g. 19,23,46,47]. They observed that weekdays such as Tuesday, Wednesday, and Thursday exhibit higher electricity demand levels, while demand on Saturdays, Sundays, and holidays is considerably lower. This distinction between workdays and non-workdays indicates the influence of societal schedules on energy usage.

Hourly statistics of electricity consumption in Ireland, as shown in Fig. 8, are primarily shaped by human daily routines. Demand typically increases in the morning as people wake up, prepare for the day, and begin their activities. It peaks again in the evening when individuals return home from work or school, engage in household activities, and require lighting and heating. Conversely, the lowest electricity demand occurs at night, when most people are asleep, and commercial and industrial activities are largely inactive.

The above analysis presents the significant influence of human behavioural patterns on electricity consumption. To account for these effects, this study incorporates day type (including day of the week and holiday status) and time of day (month and hour of the day) as representative parameters. Day type captures the impact of work schedules and holiday periods on electricity usage, while time of day reflects the influence of daily routines on consumption patterns.

3. Methodology

Machine learning techniques have been extensively applied across diverse disciplines. In time series forecasting, Recurrent Neural Networks (RNNs) and their specialized versions, such as Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRU), have been commonly employed to model future variations in time series data, such as electricity demand prediction [4,19,22,24]. These models are well-suited for capturing temporal dependencies within a complete time series. However, for datasets with incomplete historical information, techniques like linear interpolation or the use of similar past data are often employed to fill missing values [24]. Regardless of the method used, these approaches struggle to capture sudden, unexpected changes in the high-resolution data, which is the focus of this study. Such limitations can result in a loss of authenticity in the predictions—particularly when there are consecutive missing values in the dataset. For instance, in this study, the training dataset included a maximum of 99 consecutive missing data points (hours) (see Section 3.1).

Additionally, while RNN-based models are effective for short-term forecasts, where recent data points serve as strong predictors of future trends, they face significant challenges in long-term forecasting [15], which is the focus of this study. Specifically, reliance on previous predictions to inform subsequent steps leads to cumulative errors, which grow larger over longer forecast horizons. As a result, RNNs and similar models struggle to maintain accuracy in long-term predictions, making them less suitable for the scope of this research.

To overcome these limitations, this study employs a deep ANN for forecasting electricity demand. The ANN consists of an input layer, multiple hidden layers, and an output layer (see Fig. 9). Unlike RNNs, ANNs use a feature-based approach, which does not rely on sequential information or the temporal order of data points. This architecture allows ANNs to learn the complex, non-linear relationships between various input features without being influenced by prior time steps. Based on the analysis of key factors influencing electricity consumption presented in the previous section, this study identifies the most relevant features for model training. By inputting these features, the ANN is trained to model the intricate dynamics of electricity demand.

3.1. Neural network architecture and setup

Through the analysis presented above, it is clear that electricity consumption is influenced by a variety of factors, including demographic, economic, climatic, and behavioural elements. Therefore, the corresponding feature set used in this study includes population, GDP,

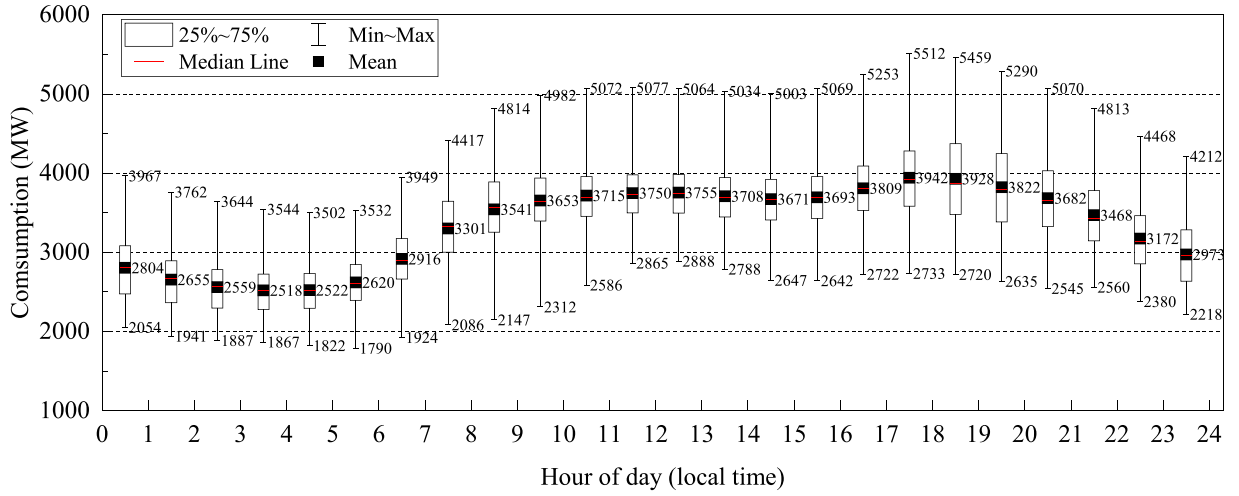


Fig. 8. Hourly statistics of electricity consumption in Ireland from 2015 to 2023.

Input Layer Hidden Layer 1 Hidden Layer 2 Output Layer

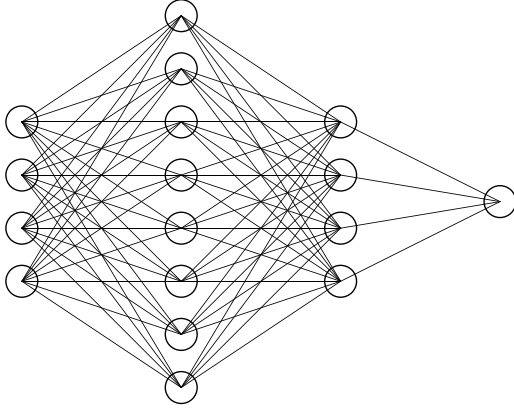


Fig. 9. Schematic diagram of Artificial Neural Network (ANN).

temperature, day type (day of the week and workday status), and time of day (month and hour of the day).

To effectively exploit these features, population, GDP, and electricity consumption, which are large in magnitude, are first rescaled by dividing electricity consumption and population by 10^3 , and GDP by 10^{12} , in order to reduce their scales and enhance the numerical stability of the neural network. Then, categorical temporal inputs, including month (12 categories), day of the week (7), hour of the day (24), and workday status (binary), are encoded independently. This approach ensures that no unintended ordinal relationships are introduced into the model, which is especially important for cyclic patterns like months, weeks or hours. Together with continuous variables such as population, GDP, and temperature (which are normalized as shown in Table 3), this results in a 47-dimensional feature vector used as input to the model.

The computations in the ANN used in this study are mathematically represented as follows:

$$h^l = \sigma(W^l h^{l-1} + b^l), \quad (1)$$

where h^l denotes the activation at layer l , with W^l and b^l representing the weights and biases at layer l . The activation function σ is applied at each hidden layer to introduce non-linearity. The final output layer provides a continuous value for electricity demand.

In this study, the above model was implemented using Keras with a Tensorflow backend [48,49]. To train the ANN effectively, the data were divided into three subsets as is standard practice in machine

learning: a training set, a validation set, and a testing set. The training set comprises hourly data from 2013 to 2021, with null data removed, resulting in 60,833 data points, which account for 78.1% of the total dataset. The validation set includes hourly data for the year 2022, also with null data removed, totalling 8428 data points, representing 10.8% of the total dataset. The testing set consists of hourly data for the year 2023, with 8673 data points after removing null values, accounting for 11.1% of the total dataset. Each dataset spans at least one full year, ensuring that the seasonal patterns in electricity demand are comprehensively captured and evaluated during training, validation, and testing.

To optimize the performance of the ANN, hyperparameters were systematically tuned. A summary of the hyperparameters involved in this neural network is presented in Table 1, with at least 14 hyperparameters included in the tuning process. Considering the complexity of exhaustively testing all possible hyperparameter combinations, this study employs Optuna [50], which leverages Bayesian optimization to efficiently identify optimal values. For each configuration of the number of hidden layers, the remaining hyperparameters were systematically adjusted, and 1000 trials were conducted. The primary objective of the Optuna optimization process was to minimize the MAPE on the validation set, as defined in Eq. (3), thereby mitigating the effects of dataset normalization. Early stopping was applied in each trial to prevent overfitting. Specifically, the validation set loss, calculated using Eq. (2), was used as the monitoring metric, and training was terminated if no improvement was observed for 50 consecutive epochs. If the validation loss continued to improve, the maximum number of training epochs was capped at 10,000. This approach enabled a thorough exploration of the model's capacity to learn complex patterns in the data while balancing computational efficiency with predictive performance.

The loss function used in training is the Mean Squared Error (MSE), defined as:

$$MSE = \frac{1}{N} \sum_{n=1}^N (y_i - \hat{y}_i)^2, \quad (2)$$

where N is the number of data sets, y_i is the actual demand, and $\hat{y}_i$ is the predicted demand. The model's performance is evaluated using the MAPE, a dimensionless metric calculated as:

$$MAPE = \frac{1}{N} \sum_{n=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right|. \quad (3)$$

3.2. Results of training, validation, and testing

The results for the training, validation, and testing sets corresponding to the optimal model from 1000 trials for each configuration of

Table 1
Summary of hyperparameters.

Hyperparameters	Data Types	Range
Number of hidden layers	Integer	[1, 6]
Number of neurones in each hidden layer	Integer	[4, 1024] (exponential variation)
Batch size	Integer	[32, 1024] (exponential variation)
Datasets (population, GDP, temperature, consumption) normalization method	Categorical	Not use, Minmaxscale, or Standardscale
Batch normalization	Boolean	Whether to use (True or False)
Optimizer	Categorical	SGD, Adam, or RMSprop
Learning rate	Float	$[10^{-5}, 10^{-1}]$ (exponential variation)
Activation function	Categorical	Rectified Linear Unit (ReLU), Exponential Linear Unit (ELU), Scaled Exponential Linear Unit (SELU), Tanh, or Sigmoid
L1L2 regularization	Boolean	Whether to use (True or False)
	Float	If True, $L1 \in [10^{-6}, 10^{-2}]$ and $L2 \in [10^{-6}, 10^{-2}]$ (exponential variation)
Dropout	Boolean	Whether to use (True or False)
	Float	If True, $droprate \in [10^{-6}, 0.99]$

Table 2
Training, validation, and testing results of ANN.

Number of hidden layers	1	2	3	4	5	6
Train error (%)	2.9483	2.6059	2.6007	2.9976	2.8849	2.5148
Validation error (%)	2.7619	2.5971	2.6838	2.7639	2.9368	2.7425
Test error (%)	2.9183	3.5548	2.8638	3.0443	3.4334	3.2626
Cross-validation error (%)	2.6238	2.5530	2.3918	2.5316	2.5406	2.5579

Table 3
Hyperparameters corresponding to the optimal model.

Hyperparameters	Value
Number of hidden layers	3
Number of neurones in each hidden layer	64, 128, 4
Batch size	256
Datasets normalization method	Temperature: Standardscale; Population, GDP and consumption: Not use
Batch normalization	False
Optimizer	Adam
Learning rate	7.94×10^{-4}
Activation function	SeLU
L1L2 regularization	True ($L1 = 1.54 \times 10^{-6}$, $L2 = 7.93 \times 10^{-6}$)
Dropout	False

the number of hidden layers, amounting to a total of 6000 trials, are summarized in Table 2, which presents the MAPE. To further evaluate model generalization and check the data splitting bias, K-fold cross-validation with $k = 5$ was also conducted, and the corresponding cross-validation error is included in the table.

Considering both validation error and test error, the ANN with three hidden layers demonstrates the most consistent and reliable performance, achieving a validation error of 2.6838% and a test error of 2.8638%. This configuration achieves a reasonable balance between training, validation, and test errors without overfitting. Therefore, the three-hidden-layer model was selected as the optimal architecture for this dataset with the corresponding hyperparameters summarized in Table 3, which will be used to forecast hourly electricity demand up to 2060.

Additionally, based on the optimal model, Figs. 10 and 11 depict the predicted and actual electricity demand for the validation and testing sets, respectively. The gaps in the curves correspond to missing actual data. While there is generally strong agreement between predicted and observed values, a noticeable peak bias occurs on Monday, January 3rd, 2022 (validation set) and Monday, January 2nd, 2023 (testing set). These days correspond to the first working days after the New Year holiday, when electricity demand typically rebounds sharply as commercial and institutional activities resume. Although the model incorporates features such as weekday and holiday indicators,

post-holiday behaviour remains highly variable and difficult to capture accurately. Despite this, the trained ANN captures most temporal patterns well.

The primary sources of error in this study can be attributed to two factors. The first is the fitting error inherent to the neural network, reflecting its limitations in approximating complex non-linear relationships in the data. The second arises from the exclusion of stochastic electricity consumption behaviours driven by random human activities, which are inherently unpredictable and not explicitly accounted for [1, 4,11,19,23,24].

As noted in [23], forecasting accuracy tends to decrease rapidly with higher resolution due to the increased variability and complexity of the data. The model presented in this study achieved a validation MAPE of 2.6838% and a test MAPE of 2.8638% at hourly resolution, which is both reasonable and acceptable given the time span and high resolution of the problem. These results suggest the model's reliability as a tool for high-resolution, long-term electricity demand forecasting.

4. Scenario descriptions

This section presents the scenarios evaluated in this study to forecast long-term electricity demand. A total of 30 scenarios are considered, derived from the combination of three key factors: population projections, economic development, and climate conditions. These scenarios provide a comprehensive framework for assessing potential variations in electricity demand over time.

4.1. Population scenarios

As urban populations and infrastructures continue to expand, the efficient management of resources has become increasingly critical [19]. Accurate population projections are essential for effective planning, enabling the optimization of resource allocation, enhancement of infrastructure efficiency, and sustainable accommodation of future electricity demand.

In this study, five key scenarios have been selected for analysis, including Medium, High Fertility, Low Fertility, No Change, and Zero-Migration (see Table 4 and Appendix). The projected population figures for each scenario are illustrated in Fig. 12. Notably, except for the High Fertility scenario, the population under the other four scenarios is expected to peak before 2060. These scenarios account for variations in fertility rates, mortality rates, and migration trends, providing a comprehensive basis for evaluating the potential impact of demographic shifts on future electricity demand.

Table 4
Population scenarios.

Scenarios	Assumptions			Description
	Fertility	Mortality	International migration	
P1: Medium	Medium ^a	Medium ^a	Medium ^a	Represents the most likely population trajectory based on current fertility, mortality, and migration trends.
P2: High fertility	High	Medium	Medium	Assumes higher birth rates than the Medium scenario, leading to a larger population.
P3: Low fertility	Low	Medium	Medium	Assumes lower birth rates than the Medium scenario, resulting in slower population growth.
P4: No change	Constant as of 2024	Constant as of 2024	Medium	Maintains current population levels with no significant growth or decline.
P5: Zero-migration	Medium	Medium	Zero from 2024	Models a scenario where there is no net migration, focusing on the effects of fertility and mortality alone.

^a Based on the mean probabilistic projections for fertility and mortality, and median probabilistic projections for net migration.

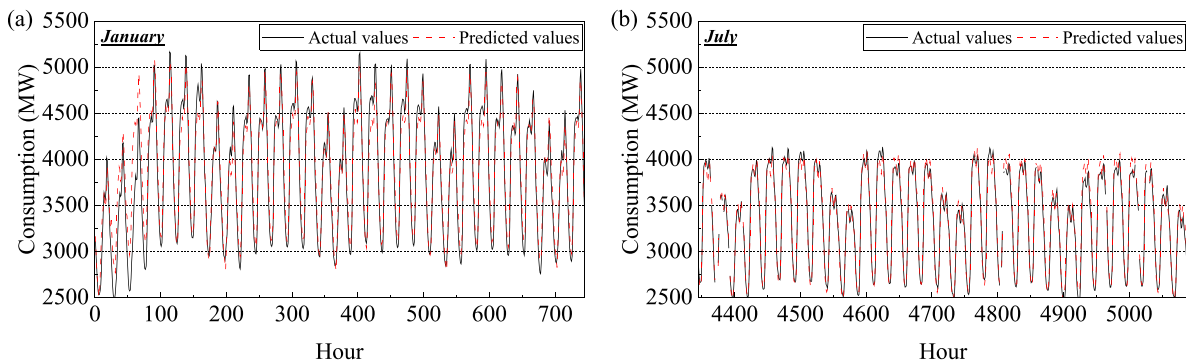


Fig. 10. Predicted and actual electricity demand for the validation sets: (a) January, 2022; (b) July, 2022.

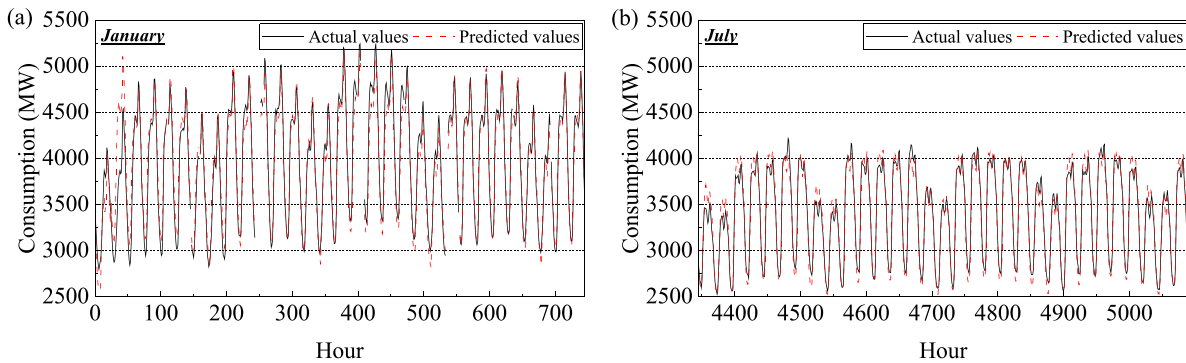


Fig. 11. Predicted and actual electricity demand for the testing sets: (a) January, 2023; (b) July, 2023.

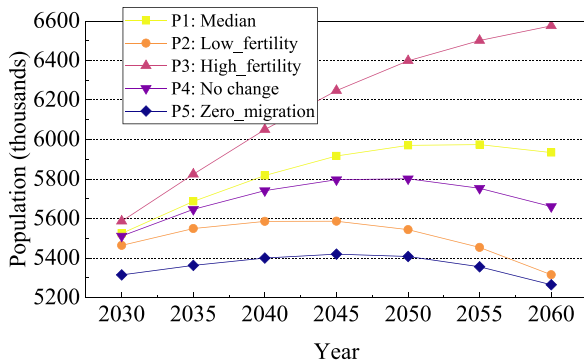


Fig. 12. Projections of population in Ireland up to 2060.

4.2. GDP scenarios

Economic activity is another critical driver of electricity demand, especially in industrial and commercial sectors. To capture potential economic trajectories, this study evaluates two distinct GDP scenarios: Baseline and Energy Transition (see Table 4, Fig. 13 and Appendix (see Table 5).

The Baseline scenario assumes a continuation of current economic trends, reflecting moderate and steady growth in GDP without significant structural changes to the energy sector. This scenario represents a business-as-usual approach, where economic development primarily drives energy demand growth across all sectors. Under this scenario, Ireland's GDP is projected to exceed 1 trillion US dollars by 2060. In contrast, the Energy Transition scenario incorporates the effects of a shift toward a low-carbon economy. It assumes accelerated investments in renewable energy, increased energy efficiency, and changes in industrial processes to align with global decarbonization goals. Under this scenario, GDP growth is moderated by the costs and structural

Table 5
GDP scenarios.

Scenarios	Description
G1: Baseline	Moderate, steady GDP growth based on OECD Real GDP Long-Term Forecast.
G2: Energy transition	GDP growth influenced by decarbonization efforts, increased energy efficiency, and structural economic changes.

Table 6
Temperature scenarios.

Scenarios	Description
T1: RCP 2.6	Represents a scenario where global warming is limited to below 2 °C by 2100, with aggressive mitigation efforts to reduce greenhouse gas emissions. This scenario assumes relatively stable temperature increases.
T2: RCP 4.5	A middle-ground scenario that assumes moderate mitigation efforts. Under this scenario, temperature increases are more pronounced than in RCP 2.6, affecting both summer cooling and winter heating demands.
T3: RCP 8.5	Reflects a business-as-usual scenario with minimal mitigation efforts, leading to significant warming by 2100. This scenario assumes drastic changes in temperature, particularly hotter summers, potentially increasing electricity demand for cooling.

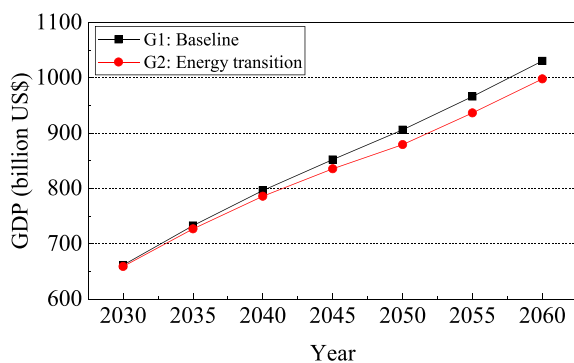


Fig. 13. Projections of GDP in Ireland up to 2060.

adjustments required to achieve a sustainable energy transition. By analysing these two scenarios, this study aims to provide a comprehensive understanding of how different economic pathways can influence future electricity demand, supporting strategic planning and policy development.

4.3. Temperature scenarios

Temperature variations significantly influence electricity demand, particularly in Ireland, where heating needs dominate energy consumption patterns. As the frequency of extreme weather events is projected to increase in the future, along with changes in weather variables due to climate change, it is crucial to examine the impact of meteorological conditions on electricity demand [16].

To account for potential climatic changes, this study evaluates three temperature scenarios, i.e., RCP 2.6, RCP 4.5, and RCP 8.5, representing varying levels of greenhouse gas concentrations (see Table 6, Fig. 14 and Appendix). According to the projections, the lowest annual temperatures under RCP 2.6, RCP 4.5, and RCP 8.5 are expected to occur around 2050, 2030, and 2040, respectively.

5. Results

We first present the results under different population scenarios, assuming the GDP scenario is G1 (Baseline) and the temperature scenario is T2 (RCP4.5). The hourly electricity demand in Ireland from 2030 to 2060 for each population scenario is depicted in Fig. 15. Despite most scenarios (except P3) reaching peak population levels before 2060, electricity consumption continues to increase annually across all scenarios. This persistent growth is primarily attributed to

economic development driving higher energy demand. Furthermore, higher population levels correlate with greater electricity consumption. Among the scenarios, P3 (high-fertility) exhibits the most significant increase in electricity demand. By 2060, the peak electricity demand under the P3 scenario could reach 6355 MW.

More specifically, Fig. 16 presents the hourly electricity demand profiles for Ireland in January of 2030, 2040, 2050, and 2060 under the P3G1T2 scenario. The curves exhibit a typical dual-peak daily pattern, with electricity demand peaking during both the morning and evening hours—the evening peak consistently higher than the morning peak, aligning the results given in Figs. 8, 10 and 11. As time progresses, electricity demand at each hour generally shows an upward trend. However, the magnitude and pattern of hourly demand are also influenced by specific temporal conditions. For example, in certain hourly windows (e.g., around hours 240, 500, and 575), the electricity demand in 2040 and 2050 exceeds that in 2060. These are mainly attributed to weekly patterns, particularly the lower demand on weekends relative to weekdays. Such temporal factors, including weekday–weekend effects, holidays, and temperature variations, can lead to deviations from the overall growth trend at specific times of day.

Fig. 17 presents the hourly electricity demand profiles for Ireland in January 2060 under different population scenarios. In the training data (see Fig. 3), population increases approximately linearly over time, which closely mirrors the trend observed in electricity consumption (see Fig. 1). As a result, the influence of population on electricity demand appears to be nearly linear in the model. Consequently, under different population scenarios, electricity demand at any given hour shows a closely positive proportion to population size.

Fig. 18 illustrates the hourly electricity demand in Ireland from 2030 to 2060 under different GDP scenarios, assuming a medium population scenario (P1) and the RCP 4.5 temperature scenario (T2). The results show that economic growth significantly drives higher electricity demand. A comparison between the two GDP scenarios suggests that efforts toward decarbonization could result in a 0.94% reduction in electricity demand by 2060.

Fig. 19 illustrates the hourly electricity demand in Ireland from 2030 to 2060 under different temperature scenarios, assuming a medium population scenario (P1) and a Baseline GDP scenario (G1). Generally, periods of high electricity demand occur during winter in Ireland due to increased heating requirements. As global temperatures rise, winter heating needs may decrease, leading to lower electricity demand. Consequently, compared to the other two scenarios, the RCP 8.5 scenario typically exhibits the lowest electricity demand during the forecast period. Additionally, as mentioned earlier, the lowest temperatures for the RCP 2.6, RCP 4.5, and RCP 8.5 scenarios are projected to occur in 2050, 2030, and 2040, respectively. These lower temperatures correspond to higher peak electricity demand in each scenario during those years, driven by increased heating needs. These peaks in electricity demand are critical for grid operators.

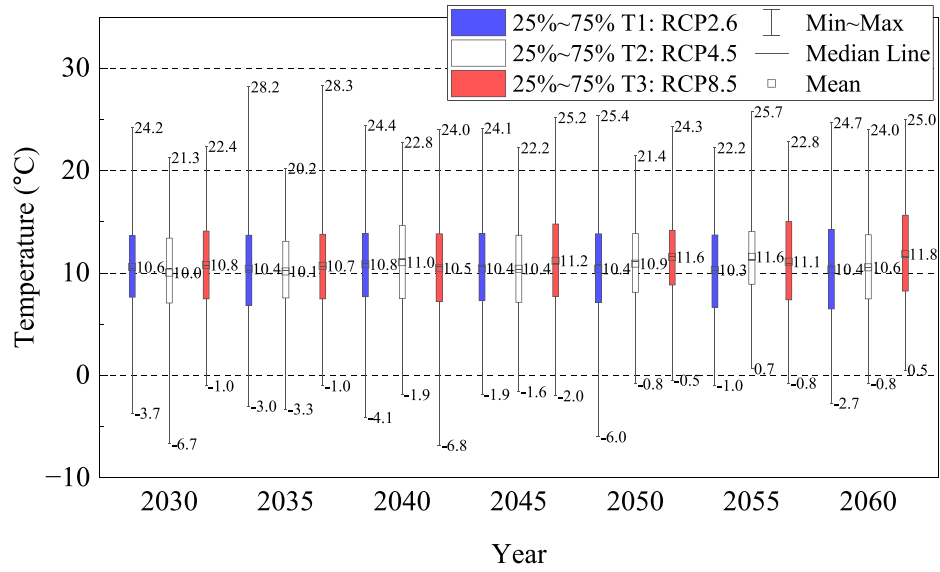


Fig. 14. Projections of temperature in Ireland up to 2060.

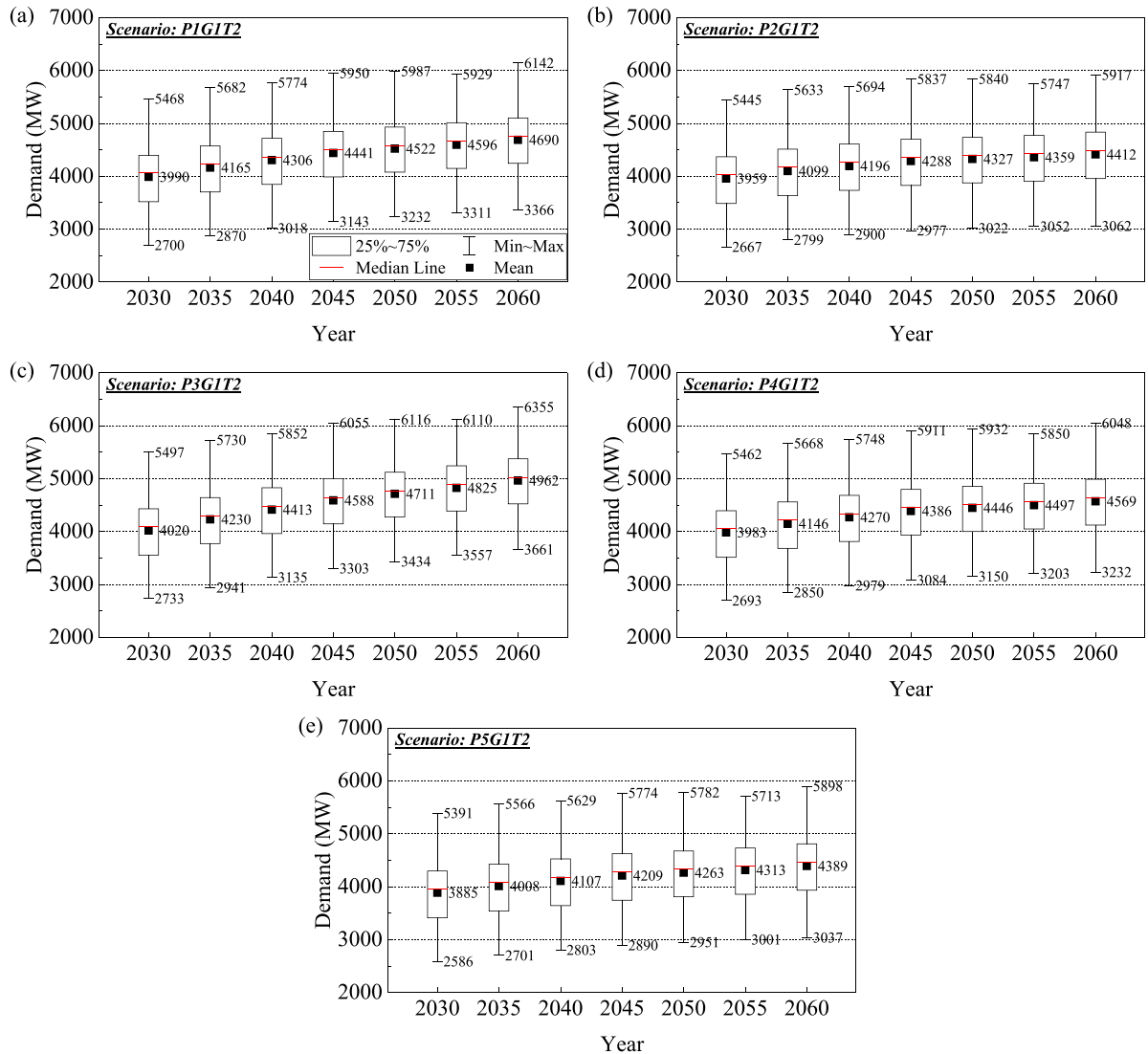


Fig. 15. Hourly electricity demand for Ireland from 2030 to 2060 under different population scenarios.

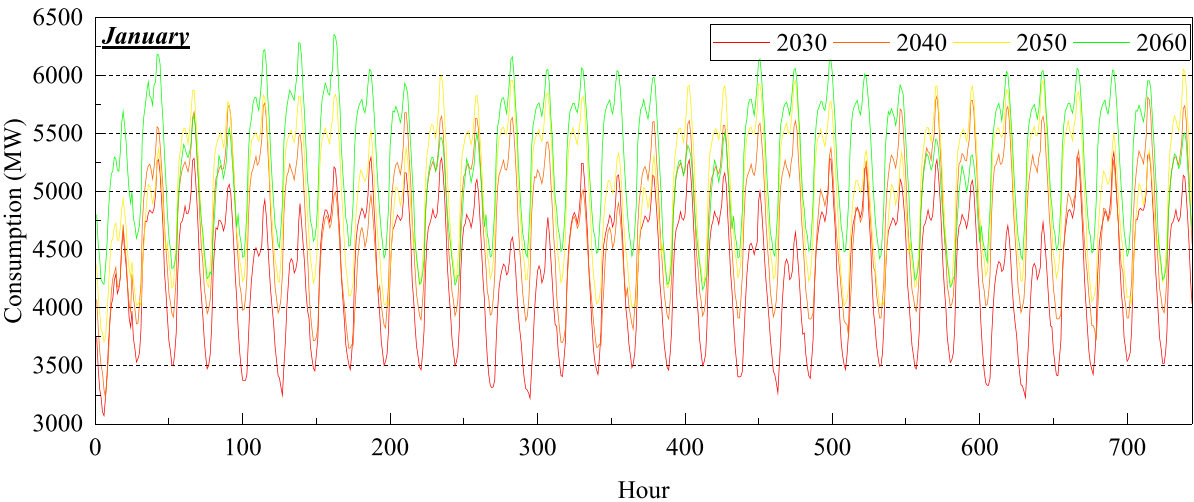


Fig. 16. Hourly electricity demand profiles for Ireland in January of 2030, 2040, 2050, and 2060 under the P3G1T2 scenario.

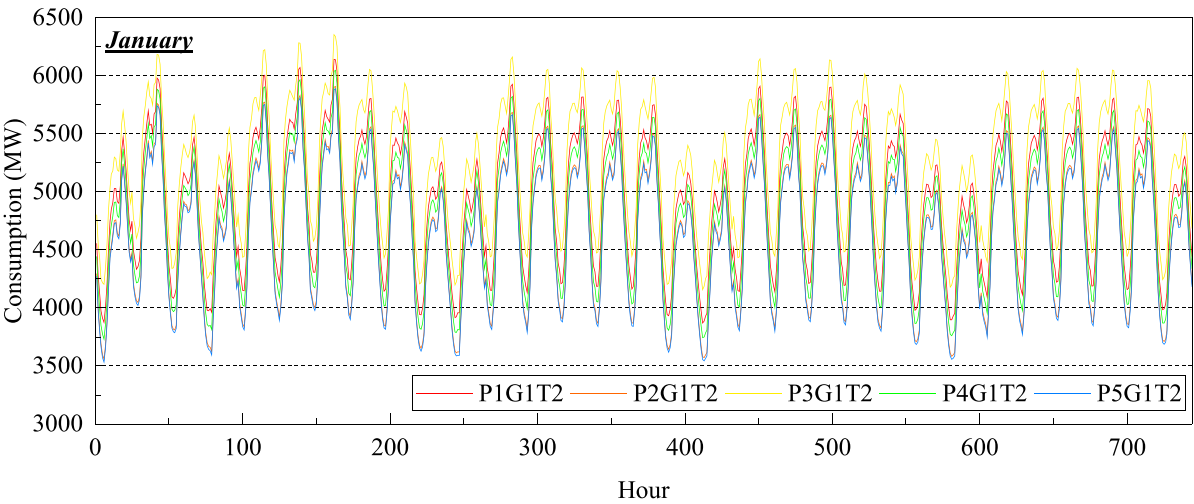


Fig. 17. Hourly electricity demand profiles for Ireland in January of 2060 under different population scenarios.

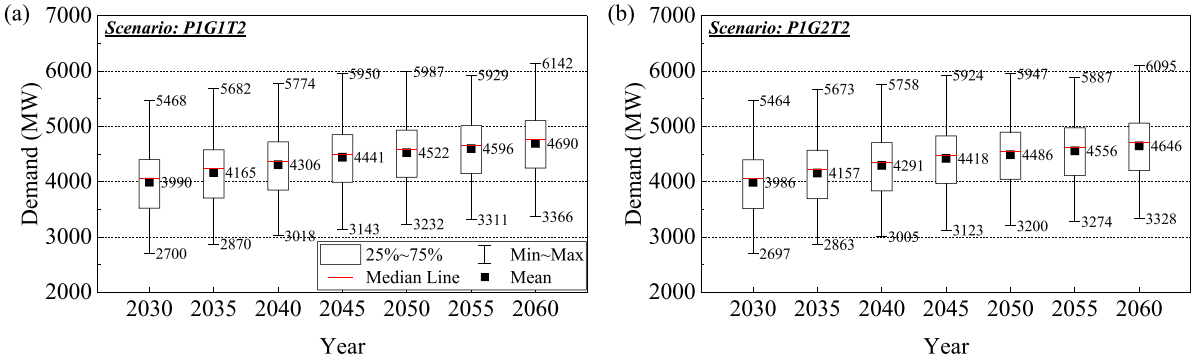


Fig. 18. Electricity demand for Ireland from 2030 to 2060 under different GDP scenarios.

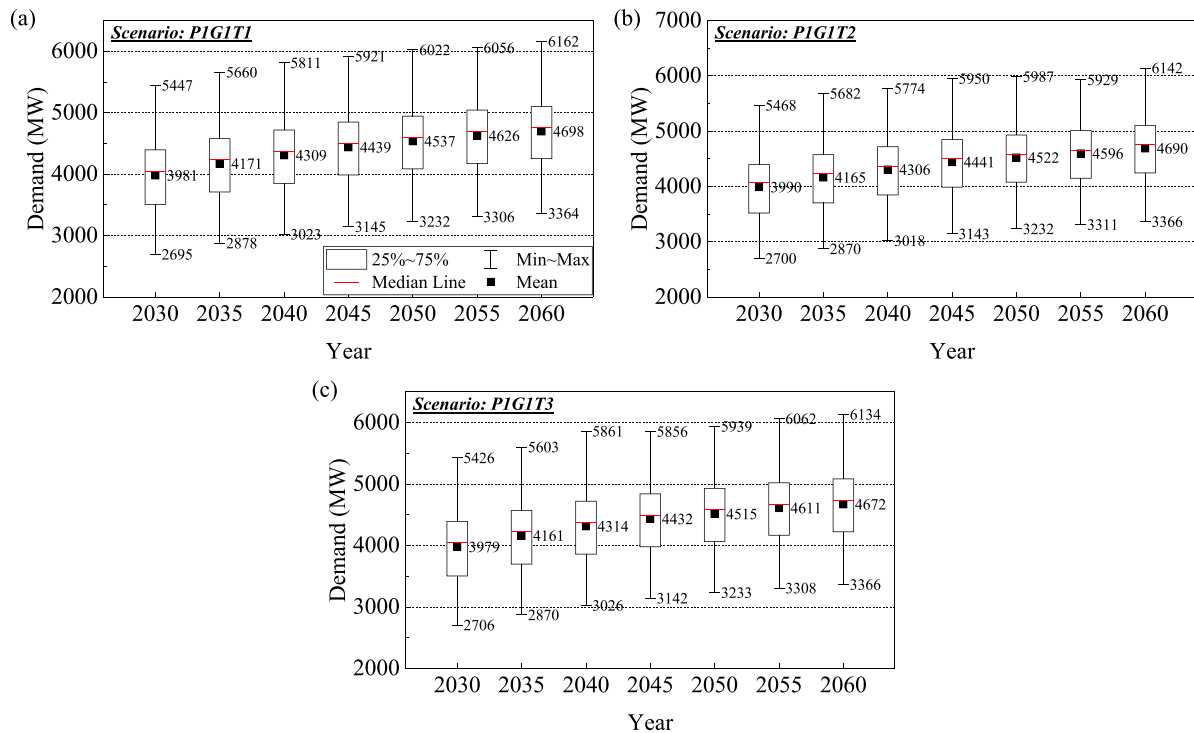


Fig. 19. Electricity demand for Ireland from 2030 to 2060 under different temperature scenarios.

6. Discussion

This study presents a framework for forecasting electricity demand, which is capable of generating long-term hourly forecasts under various socioeconomic and climatic scenarios. Compared with existing long-term forecasting models that typically focus on annual or monthly aggregates, our ANN provides hour-by-hour demand profiles up to 2060, enabling a detailed understanding of daily load variations, including peak timing, dual-peak structures, and holiday effects. Such high-resolution outputs are essential for informing future strategic decisions related to storage sizing, demand-side flexibility, and grid balancing, particularly under future conditions aligned with net-zero targets by 2050.

Another strength of the proposed method lies in its compatibility with scenario-based planning. By integrating quantified inputs for population, GDP, and temperature, the model can simulate a wide range of plausible futures. This flexibility offers an improvement over conventional energy system models that often rely on static assumptions or retrospective demand estimates, allowing for more dynamic and adaptive energy planning, and further offering greater adaptability to Ireland's net-zero targets and its evolving energy system.

Despite these strengths, it is important to acknowledge several limitations. One limitation is that the model does not explicitly include variables such as electric vehicle adoption and heat pump penetration. In spite of this, these factors are partially reflected in broader economic trends, such as GDP, which serve as proxies for underlying trends. Over the past decade, electrification of heating and transport has evolved alongside economic development in many regions, contributing to increased electricity demand. Thus, while not explicitly modelled, these influences are indirectly captured in the current ANN framework.

Another important consideration is the impact of out-of-sample forcing factors, particularly in the context of long-horizon scenario analysis. Forecasting accuracy is influenced not only by the model itself but also by the precision of the predictions for key influencing factors [46]. As such, accurately forecasting variations in population growth, economic development, and climate change is also critical for reliable electricity demand predictions.

In this study, temperature is used as a key explanatory variable to represent climatic conditions. The original data are obtained from the ERA5-Land reanalysis dataset and gridded across Ireland. These data were spatially averaged to produce a representative national temperature profile. While the spatial variability in temperature exists, the use of a spatial average is justified given Ireland's relatively small geographic area and moderate climate variability. Moreover, this approach aligns with the national-level aggregation of electricity demand and other inputs.

Despite its simplicity, the proposed ANN model achieved satisfactory accuracy and demonstrates strong transferability using publicly available datasets and a transparent modelling approach. The framework can be adapted and extended to other regions or updated as new data become available [2,24].

7. Conclusion

This study presents a high-resolution, long-term electricity demand forecast for Ireland, leveraging an ANN model under a range of socioeconomic and climate change scenarios. The historical electricity consumption patterns of Ireland were first analysed across multiple temporal scales, revealing distinct variations influenced by demographic, economic, climatic, and behavioural factors. Based on these findings, key quantitative indicators, including population, GDP, temperature, day type and time of day, were identified and incorporated into the predictive modelling framework. By integrating these indicators and applying advanced hyperparameter optimization, an optimal ANN model was developed to forecast Ireland's hourly electricity demand up to 2060. The model delivers scenario-based predictions under varying population growth rates, economic development trajectories, and climate conditions.

The forecasting results indicate a consistent annual increase in electricity demand across all population and GDP scenarios. Economic growth emerges as the primary driver, with higher population levels further amplifying electricity consumption. The climate scenarios suggest that rising global temperatures may reduce winter heating demand – particularly under the RCP 8.5 scenario – resulting in lower

electricity consumption during colder months. Conversely, increasing summer temperatures may lead to higher cooling demand, introducing consumption patterns that have historically been less prevalent in Ireland and may present challenges for the ANN model to fully capture.

Importantly, the model provides valuable hourly-level insights. The forecasted load curves exhibit typical dual-peak daily patterns, with higher evening peaks, and successfully replicate the structure of daily electricity consumption. Over time, demand increases at nearly all hours of the day, reflecting the combined influence of economic and population growth. These results highlight the model's capability not only in projecting annual trends, but also in capturing fine-grained temporal dynamics of electricity demand.

CRedit authorship contribution statement

Jin Huang: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Gregorio Iglesias:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research has received support and funding in part by the European Union's Horizon 2020 European Green Deal Research and Innovation Program (H2020-LC-GD-2020-4), grant No. 101037643 – ILIAD (Integrated Digital Framework for Comprehensive Maritime Data and Information Services).

Appendix. Data source

A.1. Electricity consumption

The electricity consumption data used in this study are obtained from the ENTSO-E Transparency Platform [30], which provides detailed and transparent information on electricity load, generation, and transmission across Europe. For the purposes of this analysis and forecasting, historical electricity load data (in MW) aggregated at the national level from 2015 to 2023 were utilized.

The original dataset has a time resolution of 30 min. To convert this to an hourly resolution, the average of two consecutive half-hour values was calculated. After cleaning the dataset by removing missing values, a total of 77,934 data points were available out of an initial 78,888.

A.2. Population

The population data used in this study, including both historical records and future projections for Ireland, are sourced from the United Nations Department of Economic and Social Affairs Population Division [51], ensuring consistency and reliability. The 2024 Revision of World Population Prospects provides probabilistic population projections, offering a comprehensive dataset that includes 13 distinct scenarios to account for uncertainty in demographic trends. From these scenarios, five key population scenarios with medium probability are selected for this study to capture a broad range of potential outcomes.

A.3. GDP

The economic growth trajectories of Ireland used in this study are derived from the OECD Real GDP Long-Term Forecast [52], ensuring a robust foundation for the analysis. This forecast is based on a comprehensive assessment of the economic climate in individual countries as well as the global economy, utilizing a combination of model-based analyses and expert judgment to provide reliable projections. The GDP indicator is expressed in USD at constant prices, adjusted for Purchasing Power Parities (PPPs), with 2015 as the base year. This ensures comparability across time and between countries, eliminating distortions caused by inflation and currency fluctuations. The use of this dataset allows for a consistent and standardized evaluation of Ireland's economic growth in both historical and future contexts.

A.4. Temperature

The historical and projected temperature data at 2-meter height are sourced from the projections provided by the [29]. These data form a reliable basis for analysing both past and future climate trends relevant to this study.

The temporal resolution of the historical temperature data is hourly, while the projected temperature data under various Representative Concentration Pathways (RCPs) are provided at a 3-h resolution. To meet the requirements of this study, the 3-hourly projections were linearly interpolated to achieve an hourly resolution. This ensures consistency in the temporal resolution of the dataset, facilitating its integration into the high-resolution forecasting framework used in this research.

Data availability

Data will be made available on request.

References

- [1] Khan AR, Mahmood A, Safdar A, Khan ZA, Khan NA. Load forecasting, dynamic pricing and DSM in smart grid: A review. *Renew Sustain Energy Rev* 2016;54:1311–22.
- [2] Ghalekhondabi I, Ardjmand E, Weckman GR, Young WA. An overview of energy demand forecasting methods published in 2005–2015. *Energy Syst* 2017;8:411–47.
- [3] Kuo P-H, Huang C-J. A high precision artificial neural networks model for short-term energy load forecasting. *Energies* 2018;11(1):213.
- [4] Shaqour A, Ono T, Hagishima A, Farzaneh H. Electrical demand aggregation effects on the performance of deep learning-based short-term load forecasting of a residential building. *Energy AI* 2022;8:100141.
- [5] Veigas M, López M, Iglesias G. Assessing the optimal location for a shoreline wave energy converter. *Appl Energy* 2014;132:404–11.
- [6] Pérez-Collazo C, Greaves D, Iglesias G. A review of combined wave and offshore wind energy. *Renew Sustain Energy Rev* 2015;42:141–53.
- [7] López M, Rodríguez N, Iglesias G. Combined floating offshore wind and solar PV. *J Mar Sci Eng* 2020;8(8):576.
- [8] Das P, Mashiat M, Iglesias G. Big data meets big wind: A scientometric review of machine learning approaches in offshore wind energy. *Energy AI* 2024;100418.
- [9] Pallonetto F, Jin C, Mangina E. Forecast electricity demand in commercial building with machine learning models to enable demand response programs. *Energy AI* 2022;7:100121.
- [10] Kamani D, Ardehali M. Long-term forecast of electrical energy consumption with considerations for solar and wind energy sources. *Energy* 2023;268:126617.
- [11] Miraki A, Parviainen P, Arghandeh R. Electricity demand forecasting at distribution and household levels using explainable causal graph neural network. *Energy AI* 2024;16:100368.
- [12] Huang J, Iglesias G. Forecasting the integration of offshore renewables into the onshore energy system up to 2050. *Renew Sustain Energy Rev* 2025;214:115523.
- [13] del Real AJ, Dorado F, Durán J. Energy demand forecasting using deep learning: applications for the French grid. *Energies* 2020;13(9):2242.
- [14] Berrisch J, Narajewski M, Ziel F. High-resolution peak demand estimation using generalized additive models and deep neural networks. *Energy AI* 2023;13:100236.
- [15] Nugalayadde A, Somaratne U, Wong KW. Predicting electricity consumption using deep recurrent neural networks. 2019, arXiv preprint [arXiv:1909.08182](https://arxiv.org/abs/1909.08182).

- [16] Apadula F, Bassini A, Elli A, Scapin S. Relationships between meteorological variables and monthly electricity demand. *Appl Energy* 2012;98:346–56.
- [17] Bunnoon P, Chalermyanont K, Limsakul C. Multi-substation control central load area forecasting by using HP-filter and double neural networks (HP-DNNs). *Int J Electr Power Energy Syst* 2013;44(1):561–70.
- [18] González-Romera E, Jaramillo-Morán M, Carmona-Fernández D. Monthly electric energy demand forecasting with neural networks and Fourier series. *Energy Convers Manage* 2008;49(11):3135–42.
- [19] Hrnjica B, Mehr AD. Energy demand forecasting using deep learning. *Smart Cities Perform Cogn, & Secur* 2020;71–104.
- [20] Kandanand K. Forecasting electricity demand in Thailand with an artificial neural network approach. *Energies* 2011;4(8):1246–57.
- [21] Rafati HHM, Jalili M, Davari H, Maknoon R. Prediction of Iran's annual electricity demand: Artificial intelligence approaches. In: 2015 11th international conference on innovations in information technology. IIT, IEEE; 2015, p. 373–7.
- [22] Son H, Kim C. A deep learning approach to forecasting monthly demand for residential-sector electricity. *Sustain*. 2020;12(8):3103.
- [23] Islam M, Che HS, Hasanuzzaman M, Rahim N. Energy demand forecasting. In: *Energy for sustainable development*. Elsevier; 2020, p. 105–23.
- [24] Román-Portabales A, López-Nores M, Pazos-Arias JJ. Systematic review of electricity demand forecast using ANN-based machine learning algorithms. *Sensors* 2021;21(13):4544.
- [25] Zhang W, Maleki A, Rosen MA, Liu J. Optimization with a simulated annealing algorithm of a hybrid system for renewable energy including battery and hydrogen storage. *Energy* 2018;163:191–207.
- [26] Sanajaoba S. Optimal sizing of off-grid hybrid energy system based on minimum cost of energy and reliability criteria using firefly algorithm. *Sol Energy* 2019;188:655–66.
- [27] Xu C, Ke Y, Li Y, Chu H, Wu Y. Data-driven configuration optimization of an off-grid wind/PV/hydrogen system based on modified NSGA-II and CRITIC-topsis. *Energy Convers Manage* 2020;215:112892.
- [28] Li J, Zhao J, Chen Y, Mao L, Qu K, Li F. Optimal sizing for a wind-photovoltaic-hydrogen hybrid system considering leveled cost of storage and source-load interaction. *Int J Hydrog Energy* 2023;48(11):4129–42.
- [29] Copernicus Climate Change Service. Climate and energy indicators for Europe from 2005 to 2100 derived from climate projections. 2021, <http://dx.doi.org/10.24381/cds.f6951a62>, (Accessed 01 December 2024).
- [30] ENTSO-EElectrifying Europe. ENTSO-E Transparency Platform. 2024, URL <https://transparency.entsoe.eu/> (Accessed 01 December 2024).
- [31] Barth R, Brand H, Meibom P, Weber C. A stochastic unit-commitment model for the evaluation of the impacts of integration of large amounts of intermittent wind power. In: 2006 international conference on probabilistic methods applied to power systems. IEEE; 2006, p. 1–8.
- [32] Keane A, Tuohy A, Meibom P, Denny E, Flynn D, Mullane A, O'Malley M. Demand side resource operation on the Irish power system with high wind power penetration. *Energy Policy* 2011;39(5):2925–34.
- [33] Welsch M, Deane P, Howells M, Gallachóir BÓ, Rogan F, Bazilian M, Rogner H-H. Incorporating flexibility requirements into long-term energy system models—a case study on high levels of renewable electricity penetration in Ireland. *Appl Energy* 2014;135:600–15.
- [34] Yao E, Samadi P, Wong VW, Schober R. Residential demand side management under high penetration of rooftop photovoltaic units. *IEEE Trans Smart Grid* 2015;7(3):1597–608.
- [35] Haikarainen C, Pettersson F, Saxén H. Optimising the regional mix of intermittent and flexible energy technologies. *J Clean Prod* 2019;219:508–17.
- [36] EnergyPLAN. EnergyPLAN. 2024, URL <https://energyplan.eu/> (Accessed 01 December 2024).
- [37] Exemplar E. PLEXOS. 2024, URL <https://www.energyexemplar.com/plexos> (Accessed 01 December 2024).
- [38] Homer Energy. HOMER. 2024, URL <https://homerenergy.com/> (Accessed 01 December 2024).
- [39] H2RES. Investigating energy transition pathways – interrelation between “power-to-X”, demand response and market coupling - INTERENERGY. 2024, URL <https://h2res.org/> (Accessed 01 December 2024).
- [40] Luz T, Moura P, de Almeida A. Multi-objective power generation expansion planning with high penetration of renewables. *Renew Sustain Energy Rev* 2018;81:2637–43.
- [41] Ara SR, Paul S, Rather ZH. Two-level planning approach to analyze techno-economic feasibility of hybrid offshore wind-solar pv power plants. *Sustain Energy Technol Assessments* 2021;47:101509.
- [42] Costoya X, DeCastro M, Carvalho D, Arguile-Pérez B, Gómez-Gesteira M. Combining offshore wind and solar photovoltaic energy to stabilize energy supply under climate change scenarios: A case study on the western Iberian peninsula. *Renew Sustain Energy Rev* 2022;157:112037.
- [43] McGrath CR, Burleyson CD, Khan Z, Rahman A, Thurber T, Vernon CR, Voisin N, Rice JS. Tell: a python package to model future total electricity loads in the United States. *J Open Source Softw* 2022;7(79):4472.
- [44] Martínez A, Iglesias G. Site selection of floating offshore wind through the levelised cost of energy: A case study in Ireland. *Energy Convers Manage* 2022;266:115802.
- [45] Ferguson R, Wilkinson W, Hill R. Electricity use and economic development. *Energy Policy* 2000;28(13):923–34.
- [46] Moral-Carcedo J, Vicéns-Otero J. Modelling the non-linear response of Spanish electricity demand to temperature variations. *Energy Econ* 2005;27(3):477–94.
- [47] Pardo A, Meneu V, Valor E. Temperature and seasonality influences on Spanish electricity load. *Energy Econ* 2002;24(1):55–70.
- [48] Chollet F, et al. Keras. 2015, (Accessed 01 December 2024) <https://keras.io>.
- [49] Abadi M, Agarwal A, Barham P, Brevdo E, Chen Z, Citro C, Corrado GS, Davis A, Dean J, Devin M, et al. Tensorflow: Large-scale machine learning on heterogeneous distributed systems. 2016, arXiv preprint [arXiv:1603.04467](https://arxiv.org/abs/1603.04467).
- [50] Akiba T, Sano S, Yanase T, Ohta T, Koyama M. Optuna: A next-generation hyperparameter optimization framework. In: *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery and data mining*. 2019.
- [51] United Nations Department of Economic Social Affairs Population Division. World Population Prospects 2024. 2024, URL <https://population.un.org/> (Accessed 01 December 2024).
- [52] OECD. Real GDP long-term forecast. 2024, URL <https://www.oecd.org/> (Accessed 01 December 2024).